

Stiff semilinear problems

Many important problems have the form

$$u'(t) = f(u) + Lu$$

where:

- f is nonlinear and non-stiff
- L is linear and stiff

Examples:

- Viscous Burgers eqn.: $u_t + uu_x = \varepsilon u_{xx}$
- KdV eqn: $u_t + uu_x + u_{xxx} = 0$
- Navier-Stokes eqns.
- Nonlinear Schrodinger:
$$iu_t = -u_{xx} \pm |u|^2 u$$

These are sometimes called stiff semilinear problems.

They're challenging because

- Explicit method $\rightarrow$ small step size
- Implicit method $\rightarrow$ nonlinear algebraic solver

We'll discuss 3 kinds of methods for efficiently solving them:

- Operator splitting
- Implicit-explicit (ImEx)
- Exponential

Operator splitting / Fractional Step methods

An operator splitting method alternates solving

$$u'(t) = f(u)$$

$$u'(t) = Lu$$

For each partial step, we either solve it exactly (if possible) or apply a time discretization.

We will write the exact or approximate solution operators as

$$e^{hf} \text{ and } e^{hL}.$$

This can be highly advantageous since one can solve the stiff part with an A-stable method and the non-stiff part with an explicit method.

Example: NLS

$$iu_t = -u_{xx} \pm |u|^2 u$$

Here $f(u) = |u|^2 u$, so we need to solve

$$iu_t = |u|^2 u \quad (1)$$

$$iu_t = \nabla^2 u \quad (2)$$

There are very efficient methods for (2). What about (1)? Observe:

$$\frac{\partial}{\partial t} |u|^2 = \frac{\partial}{\partial t} u^* u = u_t^* u + u^* u_t = 2 \operatorname{Re}(u^* u_t)$$

$$= 2 \operatorname{Re}(i u^* u |u|^2) = 2 \operatorname{Re}(i |u|^4) = 0.$$

Since $|u|^2$ is constant, (1) has a simple exact solution:

$$u(t+h) = e^{-it|u(t)|^2} u(t)$$

For this part, we don't need a discretization.

For example, Lie-Trotter / Godunov splitting:

$$u_n^* = e^{hL} u_n$$

$$u_{n+1} = e^{hf} u_n^*.$$

Even if the substeps are exact, this method is only 1st-order accurate in general. Let $f(u) = Mu$ be linear. Then the exact solution of

$$u'(t) = Mu + Lu$$

$$u(0) = u_0$$

is

$$u(h) = e^{h(M+L)} u_0$$

Lie-Trotter replaces this with

$$u_1 = e^{hL} e^{hM} u_0.$$

But

$$\begin{aligned} e^{h(M+L)} &= I + h(M+L) + \frac{h^2}{2}(M+L)^2 + \dots \\ &= I + h(M+L) + \frac{h^2}{2}(M^2 + L^2 + \underline{ML + LM}) + \dots \end{aligned}$$

$$\begin{aligned} e^{\frac{h}{2}M} e^{hL} &= (I + hM + \frac{h^2}{2}M^2 + \dots)(I + hL + \frac{h^2}{2}L^2 + \dots) \\ &= I + h(M+L) + \frac{h^2}{2}(M^2 + L^2 + \underline{2ML}) + \dots \end{aligned}$$

To get 2nd order, we need to apply M both before and after L :

$$u_1 = e^{\frac{h}{2}M} e^{hL} e^{\frac{h}{2}M} u_0$$

i.e. for our original problem:

$$u_n^* = e^{\frac{h}{2}f} u_n$$

$$u_n^{**} = e^{hL} u_n^*$$

$$u_{n+1} = e^{\frac{h}{2}f} u_n^{**}.$$

This is Strang (Strang-Marchuk) splitting.

Higher-order methods can be derived using the Baker-Campbell-Hausdorff (BCH) formula:

$$e^{hA} e^{hB} = \exp \left[h(A+B) + \frac{1}{2} h^2 C_{AB} + \frac{1}{12} h^3 (C_{AAB} + C_{BBA}) + \dots \right]$$

Where

$$C_{AB} = [A, B] = AB - BA$$

and $C_{kl \dots mn} = [k, [l, \dots [m, n] \dots]]$.

It turns out that methods with $p > 2$ must have some negative or complex coefficients. This is problematic for parabolic problems, since e.g.

$$u_t = \nabla^2 u$$

is ill-posed in reverse time.

OS methods are often used if:

- One part of the equation can be solved exactly
- One part of the equation is discretized in a complicated "legacy code"
- You want to combine a lot of different kinds of terms or discretizations
- You are lazy, you don't need much accuracy, or you don't know about other approaches.